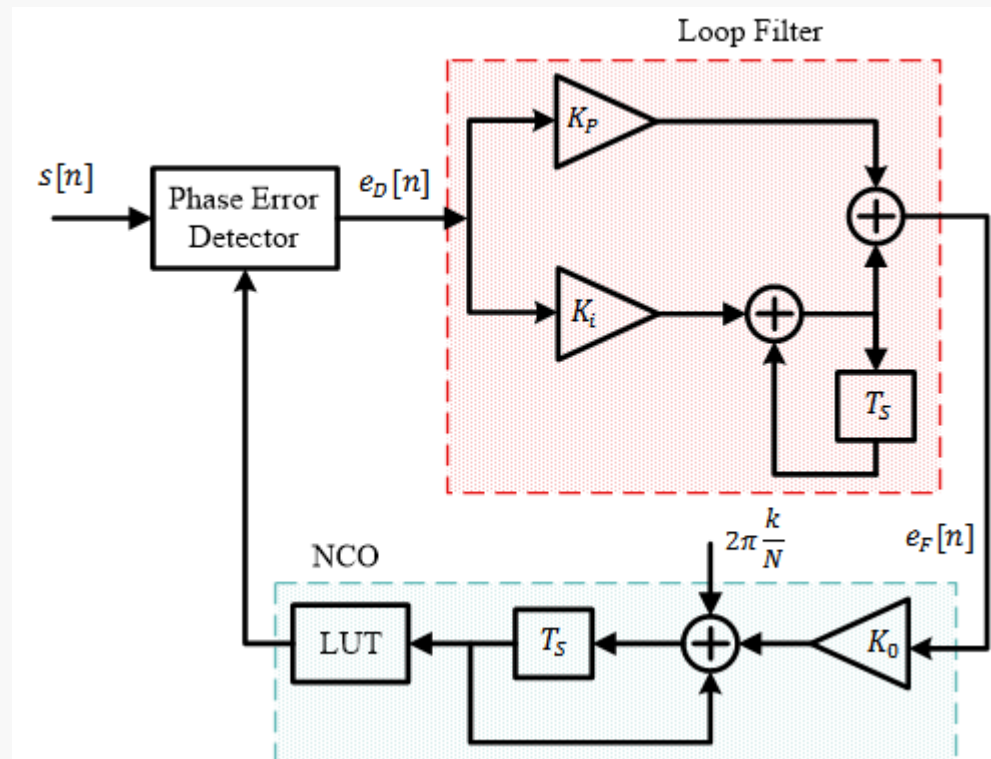


Phase Locked Loop (PLL)

Qasim Chaudhari



Introduction

- IBM Watson and Google DeepMind are the most complex computers that might run the world in a distant future



- A PLL is the simplest computer that actually runs much of the world
- A fundamental component of intelligent electronic circuits

Synchronization

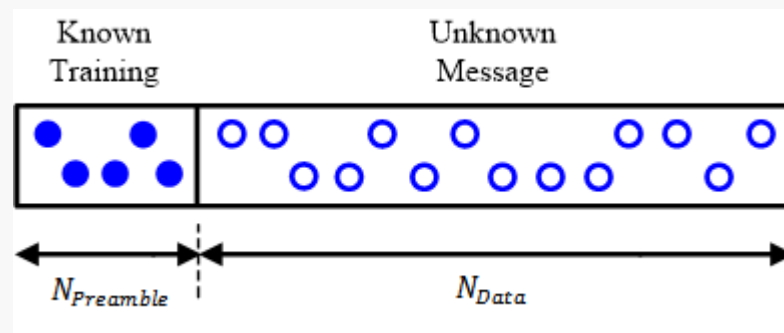
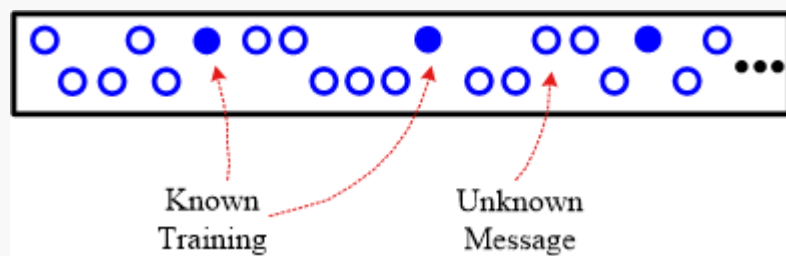
- Derived from syn (the same) and Chronos (the Greek god of time)
- Synchronizing is to make one or more things coordinate to occur in unison



- For successful operation of a digital receiver, it has to be synchronized with the incoming waveform

Synchronization

- **Timing synchronization:** sampling the Rx waveform at the proper instants of symbol boundaries
- **Carrier synchronization:** phase and frequency alignment of the local oscillator at the Rx with the incoming carrier



Data-aided: Tx inserts known symbols within the message for helping the Rx

Synchronization

Waste of resources

Decision-directed: use symbol decisions as known data to track the changes

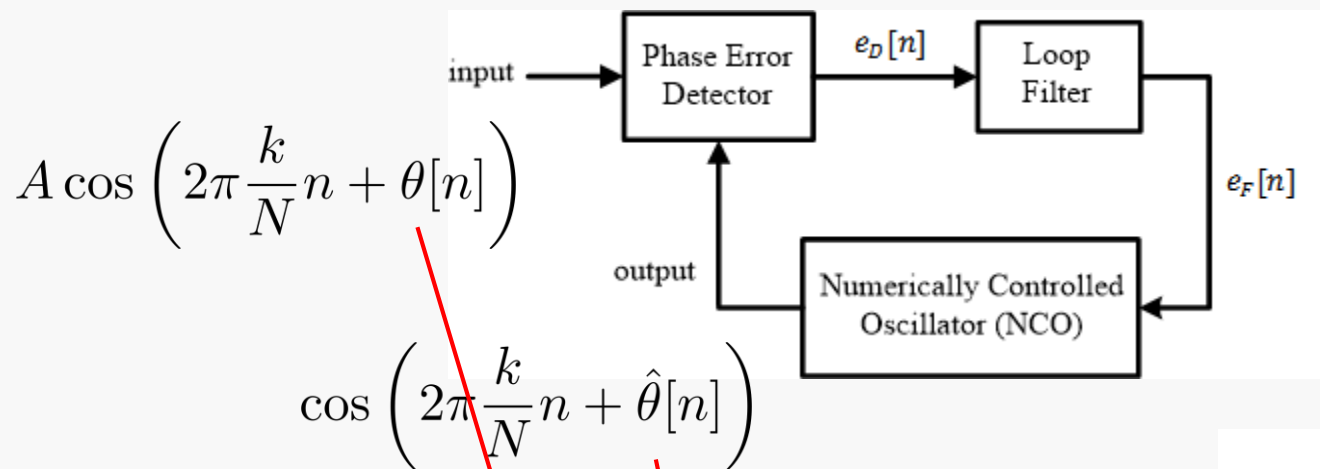
Works well only when the detector decisions are correct (a high SNR case)

Non-data-aided (blind): exploit some characteristic of the incoming signal when neither data nor decisions are available

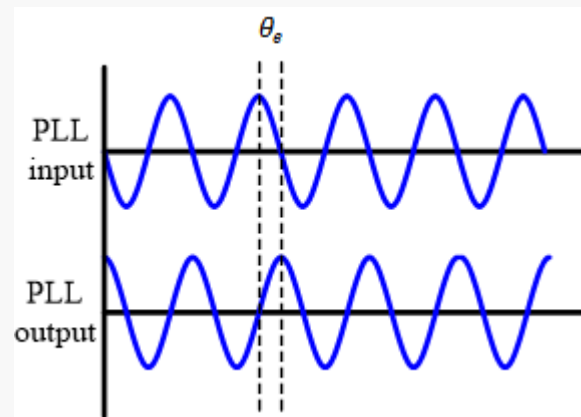
Convergence is slow but maintains the spectral efficiency

PLL

A PLL is a device used to synchronize a periodic waveform with a reference periodic waveform



phase error $\theta_e[n] = \theta[n] - \hat{\theta}[n]$

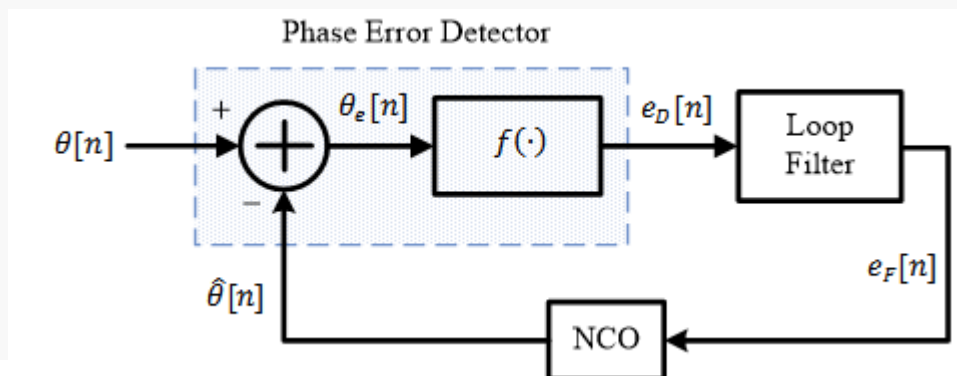
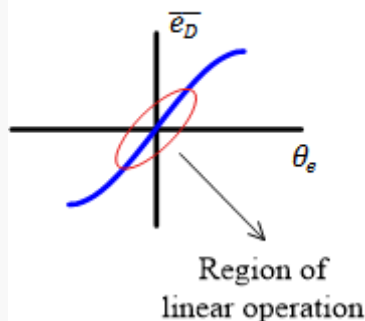


Phase Error Detector (PED)

Generates an error signal $e_D[n]$ in proportion to the phase difference $\theta_e[n]$

$$e_D[n] = f \left\{ \theta[n] - \hat{\theta}[n] \right\} = f \left\{ \theta_e[n] \right\}$$

Region of linear operation



$$f(\theta_e) \approx K_D \cdot \theta_e$$

Proportional + Integrator (PI) Loop Filter

1. Deliver a suitable control signal to the NCO to establish the dynamic performance of the loop
2. Suppress the noise and high frequency signal components

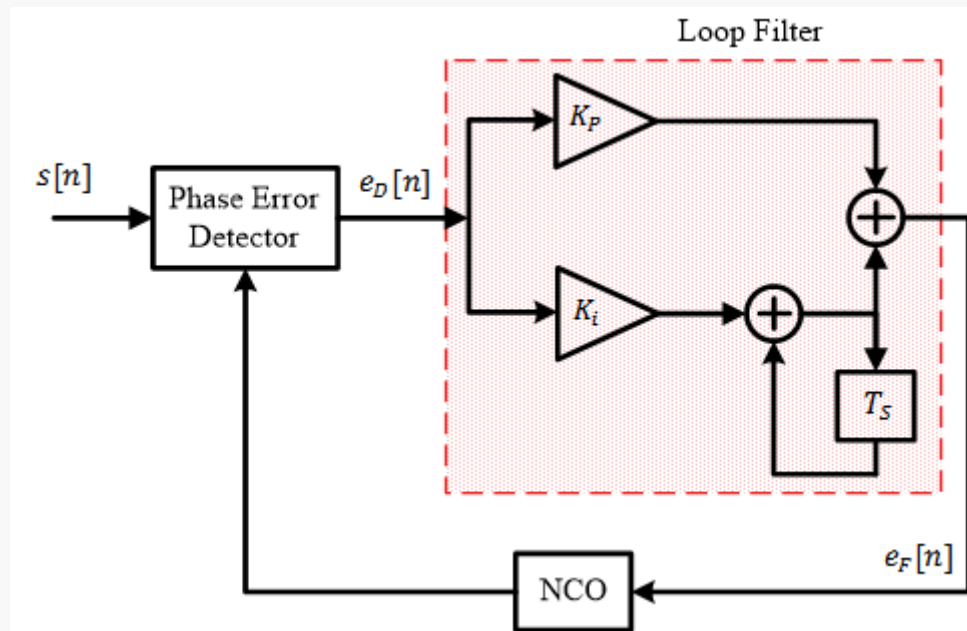
Proportional: Simple gain of K_p $e_{F,1}[n] = K_P \cdot e_D[n]$

Integrator: Integrator with a gain of K_i $e_{F,2}[n] = e_{F,2}[n-1] + K_i \cdot e_D[n]$

Combining the proportional and integrator components

$$e_F[n] = e_{F,1}[n] + e_{F,2}[n]$$

PLL with a PI Loop Filter



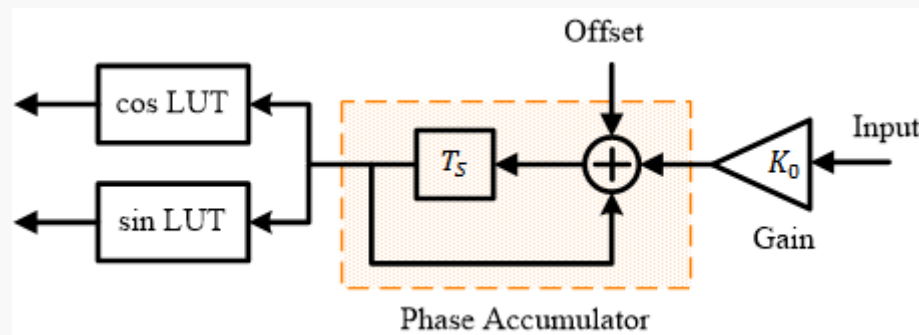
Numerically Controlled Oscillator (NCO)

Creates a discrete-time waveform whose phase is steered by digital representation of a number at its input

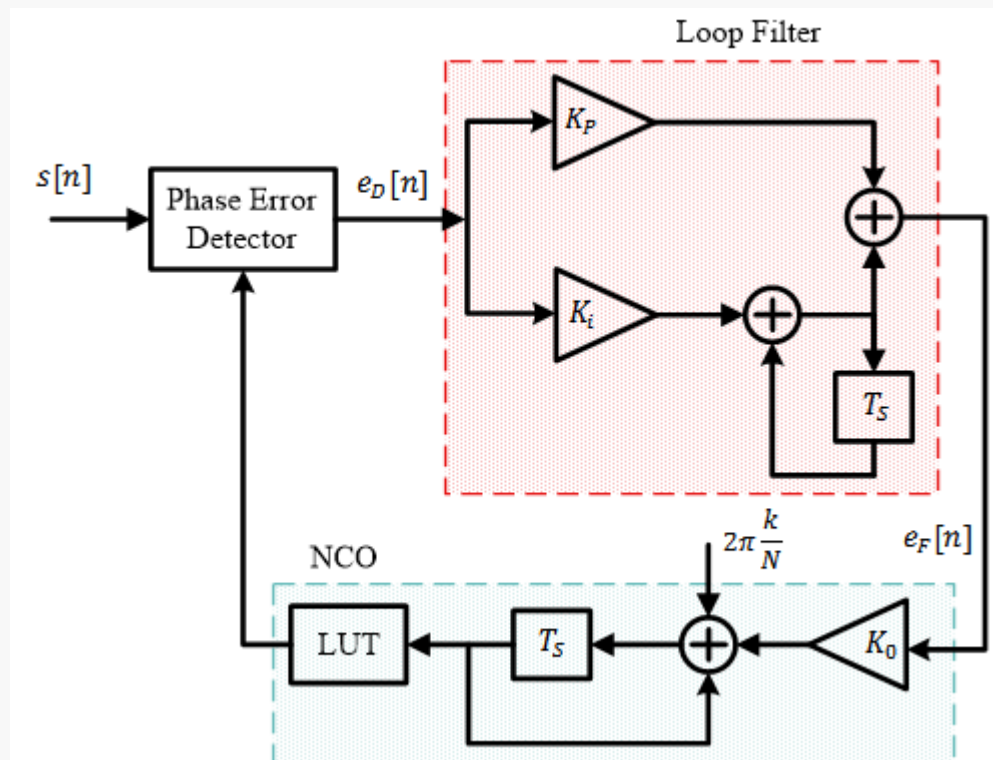
Phase accumulator:

$$\hat{\theta}[n] = \hat{\theta}[n - 1] + K_0 \cdot e_F[n - 1] \mod 2\pi$$

Look-Up Table (LUT): stores the numeric values of a desired sampled waveform (such as a sine and a cosine)



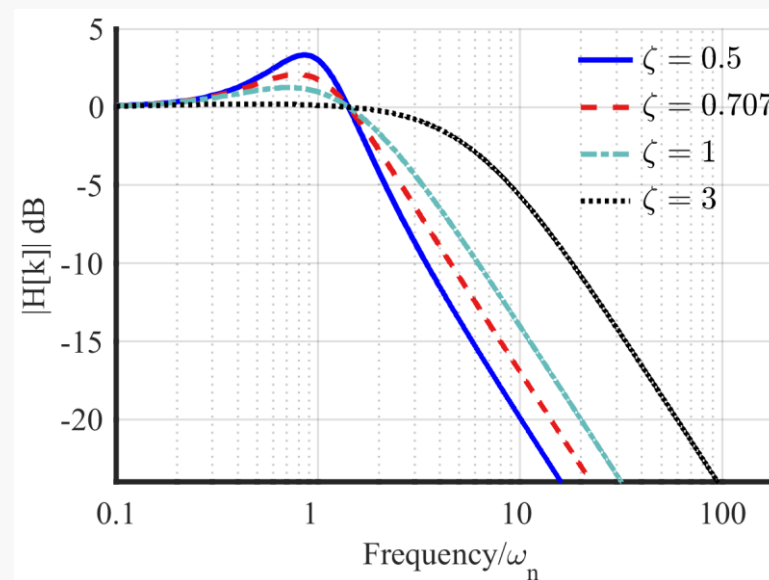
PLL with a PI Loop Filter and NCO



PLL as a Lowpass Filter

PLL in tracking mode acts as a lowpass filter

In this role, the natural frequency ω_n can be considered as a coarse measure of the loop bandwidth



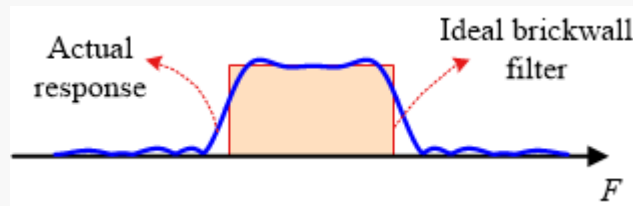
Loop Bandwidth B_n

Loop frequency response depends on damping factor ζ for the same ω_n

Phase acquisition process exhibits an oscillatory behaviour at the start before settling towards a steady state



Loop bandwidth (actually known as equivalent noise bandwidth) B_n



$$B_n = \frac{\omega_n}{2} \left(\zeta + \frac{1}{4\zeta} \right)$$

PLL Design

Choose the damping factor ζ

- Large ζ results in no overshoots but long convergence time
- Small ζ exhibits relatively fast convergence but damped oscillations
- A good balance between the two is achieved with $\zeta = 1/\sqrt{2}$

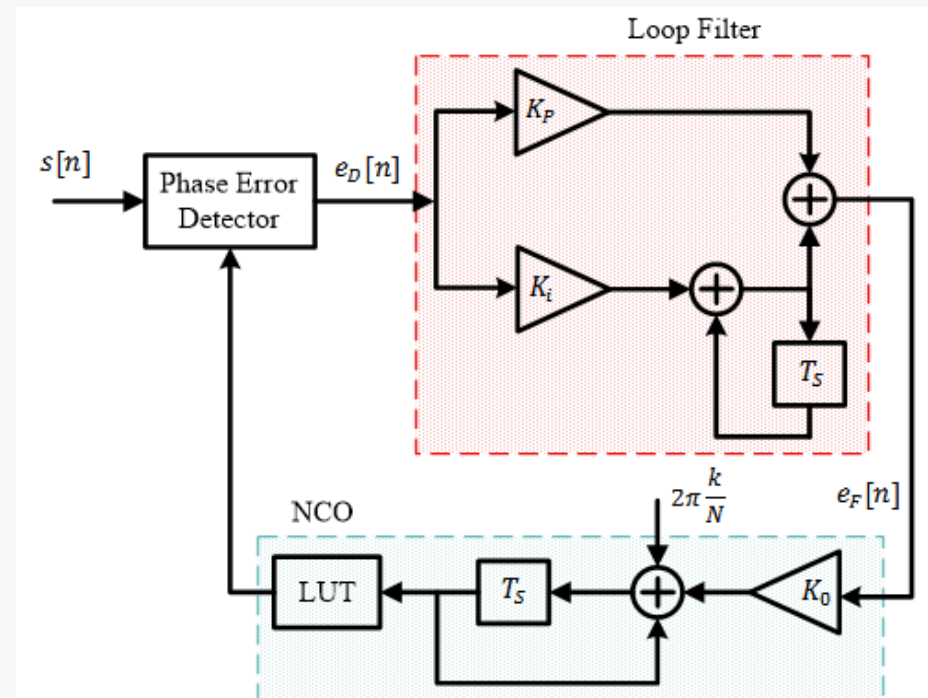
Choose the loop bandwidth B_n

- Small B_n filters out most of the noise and frequencies in the stopband
- Large B_n can track fast phase variations (higher frequencies) but allow more noise enter the loop
- A value of around 1/100 of the input rate serves as a good starting point

PLL Design

Having chosen a PI loop filter, there are four constants that need to be determined

- **NCO gain K_0 :** In a discrete-time system, K_0 can be easily fixed to a suitable value, say 1
- **Phase error detector gain K_D :** depends on structure and expression of the phase error detector

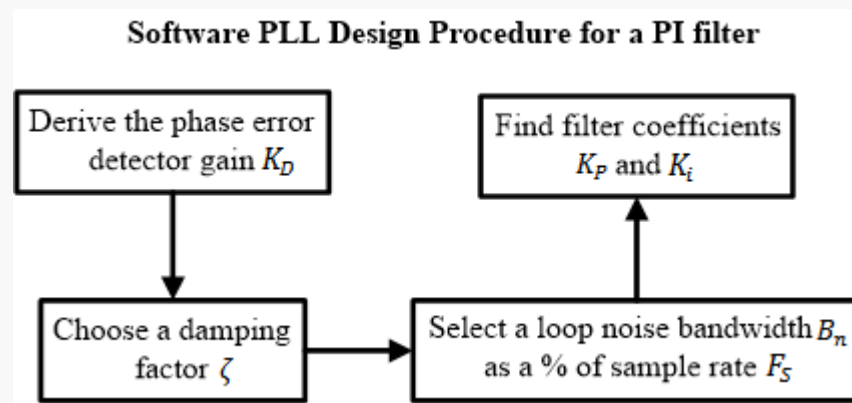


PLL Design

From control systems theory $\theta_n = \omega_n \frac{T_S}{2} = \frac{B_n T_S}{\zeta + \frac{1}{4\zeta}}$

$$K_P = \frac{1}{K_D K_0} \cdot \frac{4\zeta\theta_n}{1 + 2\zeta\theta_n + \theta_n^2}$$

$$K_i = \frac{1}{K_D K_0} \cdot \frac{4\theta_n^2}{1 + 2\zeta\theta_n + \theta_n^2}$$



[1] Michael Rice, 'Digital Communications – A Discrete-Time Approach', Prentice Hall 2009

GNU Radio

- In block 'Polyphase Clock Sync', system is critically damped ($\zeta = 1$)
- Since loop bandwidth is very small, ignore $2\zeta\theta_n + \theta_n^2$

$$K_P = 4\theta_n$$

$$K_i = 4\theta_n^2$$

- We can write

$$K_i = \frac{K_P^2}{4}$$

- From here

$$\text{'alpha'} = \text{'gain'}$$

$$\text{'beta'} = \frac{\text{'gain'}^2}{4}$$

Example

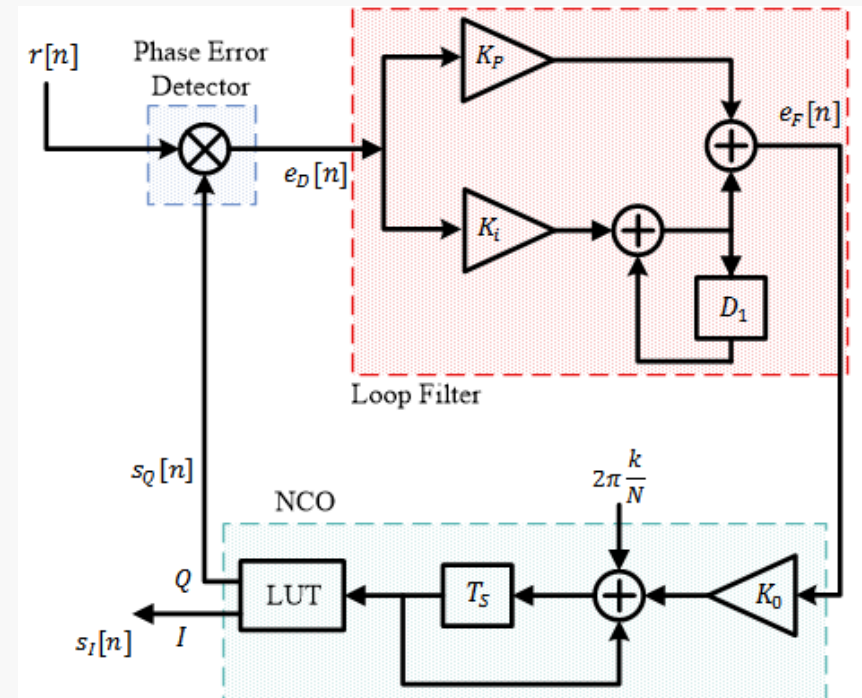
Input

$$r[n] = A \cos \left(2\pi \frac{1}{15} n + \theta[n] \right)$$

Output

$$s_I[n] = \cos \left(2\pi \frac{k}{N} n + \hat{\theta}[n] \right)$$

$$s_Q[n] = -\sin \left(2\pi \frac{k}{N} n + \hat{\theta}[n] \right)$$

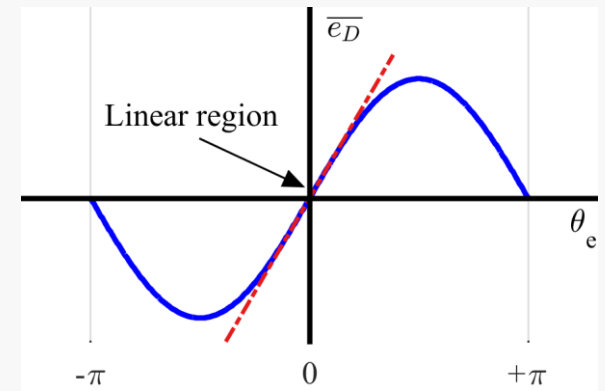


Example

Phase error detector output

$$e_D[n] = -\sin\left(2\pi\frac{k}{N}n + \hat{\theta}[n]\right) \cdot A \cos\left(2\pi\frac{k}{N}n + \theta[n]\right) \approx \frac{A}{2} \sin \theta_e[n]$$

Clearly, the detector gain is $K_D = \frac{A}{2}$



Choose a damping factor $\zeta = 0.707$ and loop noise bandwidth $B_n T_S = 0.05$
(5% of the sample rate)

$$K_P = 0.2667$$

$$K_i = 0.0178$$

Example

